DESIGN AND DEVELOPMENT OF A CLASSICAL CRYPTOGRAPHY TOOLKIT

Final Year Engineering Mini-Project Report (2026–27)
Course: Network Security | Branch: Electronics and Computer Engineering (ECM)

Academic Year: 2026–2027	**Department:** Electronics and Computer Engineering
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1. PROJECT OBJECTIVES

This mini-project delivers a mathematically validated, modular Classical Cryptography Toolkit designed for Network Security education. The project encompasses eight classical ciphers categorized into Substitution (Caesar, Monoalphabetic, Playfair, Hill, Vigenère, One-Time Pad) and Transposition (Rail Fence, Columnar Transposition). A Tkinter graphical user interface offers 13 structured menu options, execution timing, and safe UTF-8 file operations.

2. SUPPORTED ALGORITHMS & IMPLEMENTATION RULES

Cipher Name	Category	Mathematical Rule / Principle	Key Format
Caesar Cipher	Substitution	C = (P + k) mod 26. Verified: 'NETWORK SECURITY' + 5 -> 'SJYBTWP XJHZWNYD'.	Integer shift k
Monoalphabetic	Substitution	Bijection mapping standard A-Z to unique 26-letter substitution alphabet.	26 unique letters or keyword
Playfair Cipher	Substitution	5x5 matrix digraphs (J=I). Automatic filler 'X'/'Q' insertion and odd padding.	Alphabetic keyword
Hill Cipher	Substitution	Modular linear algebra: C = (P * K) mod 26. Enforces gcd(det(K), 26) = 1.	2x2 or 3x3 matrix
Vigenère Cipher	Substitution	Polyalphabetic shift: C_i = (P_i + K_j) mod 26. Preserves spaces/punctuation.	Alphabetic keyword
One-Time Pad	Substitution	Information-theoretic secrecy. CSPRNG secrets generation; key length = text length.	Random A-Z keystream
Rail Fence	Transposition	Diagonal zigzag traversal across k rails. Exact character round-trip restoration.	Integer depth >= 2
Columnar Trans.	Transposition	Matrix rows read by alphabetical key column order. Handles irregular grids.	Keyword or numeric order

3. AUTOMATED TESTING AND BENCHMARK RESULTS

Automated unit and integration testing was executed using pytest 9.1.1 under Python 3.14.3. A total of **58 test cases were executed, achieving a 100% pass rate in 0.25 seconds** with zero errors or failures. All edge cases including empty inputs, single characters, repeated letters, non-invertible matrices, and irregular grids passed successfully.

Algorithm	Input Len	Cipher Len	Enc Time (ms)	Dec Time (ms)	Round-Trip
Caesar Cipher	45	45	0.0190	0.0176	Passed
Monoalphabetic	45	45	0.0359	0.0237	Passed
Playfair Cipher	45	42	0.1386	0.0984	Passed
Hill Cipher (2x2)	45	42	0.1244	0.1805	Passed
Vigenère Cipher	45	45	0.0393	0.0349	Passed
One-Time Pad	45	45	0.0528	0.0440	Passed
Rail Fence Cipher	45	45	0.0273	0.0424	Passed
Columnar Transposition	45	45	0.0347	0.0377	Passed

4. CYBERSECURITY AND PEDAGOGICAL NOTICE

Important Disclaimer: Classical ciphers are historical educational instruments intended to demonstrate foundational principles of cryptography, confusion, diffusion, and cryptanalysis. They do not resist modern computational cryptanalytic attacks and must NEVER be deployed to secure confidential or sensitive real-world communications.